

B28: SQL SuperHero Program : Session 24

1 message

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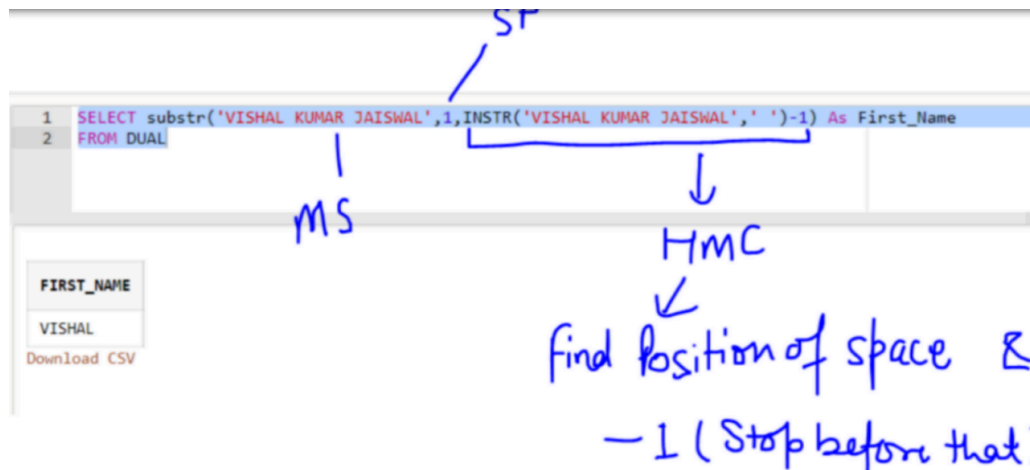
24 March 2025 at 03:50

INSTR/SUBSTR Continue:

Take any example, suppose my name :

VISHAL KUMAR JAISWAL

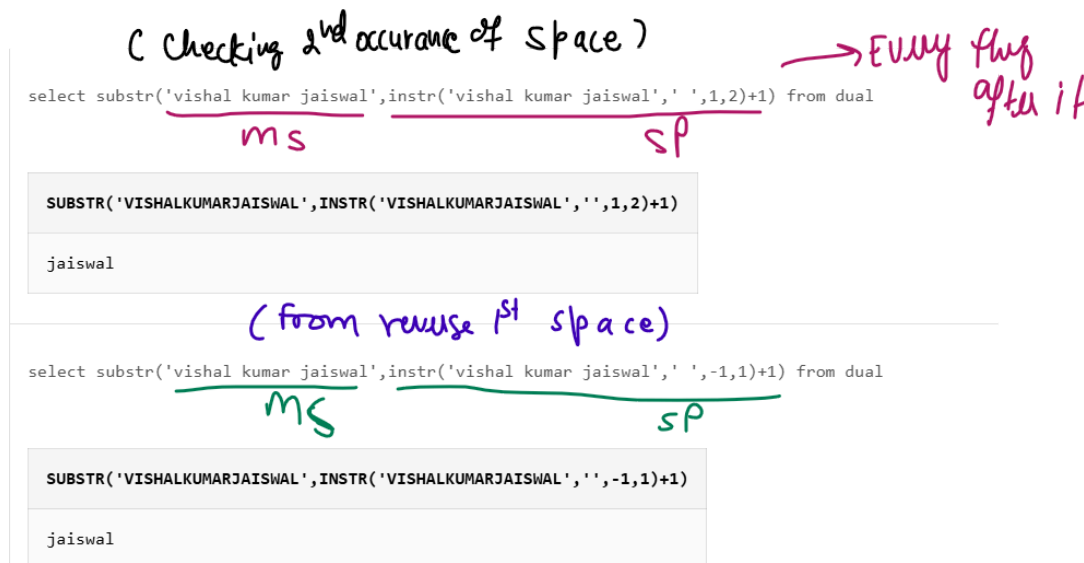
1. I want first name --> start from 1 and stop when I found first time space



The screenshot shows a SQL query in a text editor: `SELECT substr('VISHAL KUMAR JAISWAL',1,INSTR('VISHAL KUMAR JAISWAL',' ')-1) As First_Name FROM DUAL`. Handwritten blue annotations include 'MS' pointing to the space character in the string, 'ST' pointing to the space character in the INSTR function, and 'HMC' pointing to the '-1' in the INSTR function. Below the query, a result table shows 'FIRST_NAME' with the value 'VISHAL'. A note below the table says 'find position of space & -1 (Stop before that)'.

2. I want a last name. --> There is 2 options:

1. start after 2nd space and give me everything
2. Find the space from reverse and after that everything



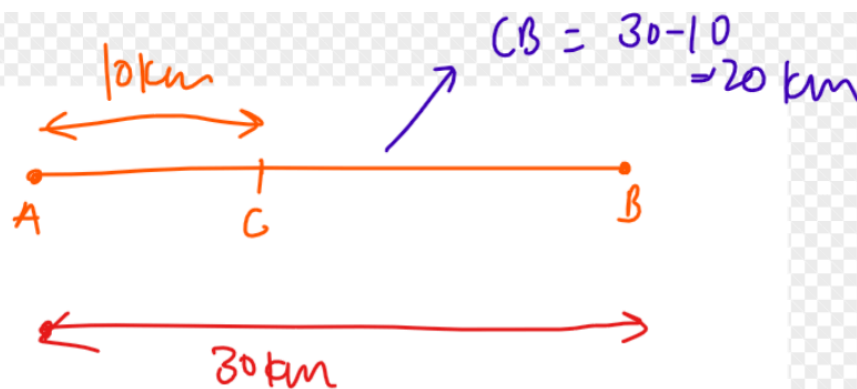
The first screenshot shows a SQL query: `select substr('vishal kumar jaiswal',instr('vishal kumar jaiswal',' ',1,2)+1) from dual`. Handwritten pink annotations include 'ms' pointing to the first space, 'sp' pointing to the second space, and a note '(Checking 2nd occurrence of space)'. A red arrow points to the right with the text 'Every thing after it'. The result table shows 'SUBSTR('VISHALKUMARJAISWAL',INSTR('VISHALKUMARJAISWAL',' ',1,2)+1)' and the value 'jaiswal'.

The second screenshot shows a SQL query: `select substr('vishal kumar jaiswal',instr('vishal kumar jaiswal',' ',-1,1)+1) from dual`. Handwritten green annotations include 'ms' pointing to the first space and 'sp' pointing to the last space. A note above the query says '(from reverse 1st space)'. The result table shows 'SUBSTR('VISHALKUMARJAISWAL',INSTR('VISHALKUMARJAISWAL',' ',-1,1)+1)' and the value 'jaiswal'.

3. I want Middle Name --> here is the trick. we know middle name means start after 1st space and stop before 2nd space

- After 1st space means --> `instr('VISHAL KUMAR JAISWAL',' ')+1`
- Before 2nd space means --> `instr('VISHAL KUMAR JAISWAL',' ',1,2)-1`

These both are from the start of the string.



```
select instr('vishal kumar jaiswal',' ') first_space, instr('vishal kumar jaiswal',' ',1,2) second_space from dual
```

FIRST_SPACE	SECOND_SPACE
7	13

→ we want $13 - 7 = 6$ character

```

select substr('vishal kumar jaiswal',1,instr('vishal kumar jaiswal',' ')-1) first_name,
       substr('vishal kumar jaiswal', instr('vishal kumar jaiswal',' ',1,2)+1) last_name ,
       substr('vishal kumar jaiswal', instr('vishal kumar jaiswal',' ')+1, instr('vishal kumar jaiswal',' ',1,2)-instr('vishal kumar jaiswal',' ')-1) m
from dual

```

FIRST_NAME	LAST_NAME	MIDDLE_NAME
vishal	jaiswal	kumar

ms

sp (after 1st space)

```
create table empdata(name varchar2(30))
```

```

insert into empdata values ('Mr. Neil Johnson');
insert into empdata values ('Mrs. sweta prasar');
insert into empdata values ('Miss. douglas tahir');
insert into empdata values ('Ms. angelina jolly');

```

```

select name, substr(name, 1,instr(name,'.')) from empdata; -- Title
select name, substr(name, instr(name,'.')+1, instr(name,'.',1,2)-instr(name,'.')->1) from empdata; -- Firstname
select name, substr(name, instr(name,'.',1,2) ) from empdata; -- LastName

```

ms sp find space & stop before it

ms sp HMC (find position of . & stop)

```

1 select name, substr(name,1,instr(name,' ')) title, substr(name,1,instr(name,' ')-1) as totle from empdata
2
3

```

Script Output x Query Result 2 x

SQL | All Rows Fetched: 4 in 0.009 seconds

	NAME	TITLE	TOTLE
1	Mr. Neil Johnson	Mr.	Mr.
2	Mrs. sweta prasar	Mrs.	Mrs.
3	Miss. douglas tahir	Miss.	Miss.
4	Ms. angelina jolly	Ms.	Ms.

See the above sql. Any method, whatever you think you can use, either find the position of dot (.) or find the space and stop before that. Last name is easy, just find the 2nd space and take everything, or take first space from back.

Partitioning space from last

Position of 2nd Space from 1st

```

1 select name, substr(name,instr(name,' ',1,2)+1) last_name, substr(name,instr(name,' ',-1,1)+1) as lastname from empdata

```

Script Output x Query Result x

SQL | All Rows Fetched: 4 in 0.004 seconds

	NAME	LAST_NAME	LASTNAME
1	Mr. Neil Johnson	Johnson	Johnson
2	Mrs. sweta prasar	prasar	prasar
3	Miss. douglas tahir	tahir	tahir
4	Ms. angelina jolly	jolly	jolly

Few More Question we discussed :

Question

Write a query in SQL that will give the first employee name starting with a p letter and h letter and the remaining employee names in ascending order. Table employee having employee name column.

Understand here the conditional construct. If name start from P then it should come 1st, If name starts from H then it should be after that. If then else means case

```
select * from hr.employees
order by case when FIRST_NAME like 'P%' then 'AAA' when FIRST_NAME like 'H%' then 'AAB' else First_name end
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SAL
157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500
202	Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6000
152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000
150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000
144	Peter	Vargas	PVARGAS	650.121.2004	09-JUL-06	ST_CLERK	2500
122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900
169	Harrison	Bloom	HBLOOM	011.44.1343.829268	23-MAR-06	SA_REP	10000

```
select * from hr.employees
order by case when instr(FIRST_NAME,'P') =1 then 'AAA' when instr(FIRST_NAME,'H') =1 then 'AAB' else First_name end
```

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT
157	Patrick	Sully	PSULLY	011.44.1345.929268	04-MAR-04	SA_REP	9500	.35
202	Pat	Fay	PFAY	603.123.6666	17-AUG-05	MK_REP	6000	-
152	Peter	Hall	PHALL	011.44.1344.478968	20-AUG-05	SA_REP	9000	.25
150	Peter	Tucker	PTUCKER	011.44.1344.129268	30-JAN-05	SA_REP	10000	.3
144	Peter	Vargas	PVARGAS	650.121.2004	09-JUL-06	ST_CLERK	2500	-
122	Payam	Kaufling	PKAUFLIN	650.123.3234	01-MAY-03	ST_MAN	7900	-
169	Harrison	Bloom	HBLOOM	011.44.1343.829268	23-MAR-06	SA_REP	10000	.2

=====

Write a Query to Update The gender of table. Update Male to Female and Female to Male. (Most asking question in Interview)

Understand the logic again. If gender = Male then change to Female and If gender = Female then change to Male. If Then else means case decode.

```
select * from users
```

USER_ID	FIRST_NAME	LAST_NAME	GENDER
1	John	Doe	male
2	Jane	Smith	female
3	Alice	Johnson	female
4	Bob	Brown	male
5	Charlie	Davis	male
6	Eve	White	female

this is input table

6 rows selected.

Decode or Case anything

```
update users set gender = decode(gender,'male','female','female','male')
```

6 row(s) updated.

✓

```
Update users set gender = case when gender = 'male' then 'female'
                                when gender = 'female' then 'male' end
```

6 row(s) updated.

✓

=====

```
create table customer_order (Customer_Name varchar2(50), order_id varchar2(10), status varchar2(20));
Insert Into Customer_Order Values ('John','J1','DELIVERED');
Insert Into Customer_Order Values ('John','J2','DELIVERED');
Insert Into Customer_Order Values ('John','J3','DELIVERED');
Insert Into Customer_Order Values ('David','D1','SUBMITTED');
Insert Into Customer_Order Values ('David','D2','DELIVERED');
Insert Into Customer_Order Values ('David','D3','CREATED');
Insert Into Customer_Order Values ('Smith','S1','SUBMITTED');
Insert Into Customer_Order Values ('Smith','S2','CREATED');
Insert Into Customer_Order Values ('Smith','S3','SUBMITTED');
Insert Into Customer_Order Values ('KRISH','K1','CREATED');
Insert Into Customer_Order Values ('Avnee','A1','DELIVERED');
Insert Into Customer_Order Values ('Avnee','A2','DELIVERED');
Insert Into Customer_Order Values ('Avnee','A3','DELIVERED');
Insert Into Customer_Order Values ('Avnee','A4','SUBMITTED');
Insert Into Customer_Order Values ('Sruti','SR1','SUBMITTED');
Insert Into Customer_Order Values ('Sruti','SR2','CREATED');
Insert Into Customer_Order Values ('Sruti','SR3','SUBMITTED');
Insert Into Customer_Order Values ('Michael','M1','DELIVERED');
Insert Into Customer_Order Values ('Michael','M2','DELIVERED');
Insert Into Customer_Order Values ('Michael','M3','DELIVERED');
Insert Into Customer_Order Values ('Zoe','Z1','CREATED');
Insert Into Customer_Order Values ('Harris','H1','SUBMITTED');
Insert Into Customer_Order Values ('Harris','H2','DELIVERED');
Insert Into Customer_Order Values ('Harris','H3','CREATED');
Insert Into Customer_Order Values ('Emmy','E1','DELIVERED');
Insert Into Customer_Order Values ('Emmy','E2','DELIVERED');
```

```

Insert Into Customer_Order Values ('Emmy','E3','DELIVERED');
Insert Into Customer_Order Values ('Hanna','H1','SUBMITTED');
Insert Into Customer_Order Values ('Hanna','H2','CREATED');
Insert Into Customer_Order Values ('Hanna','H3','SUBMITTED');

```

We expect the output like :

1. if all orders for a customer is delivered then print "completed"
2. if any one is delivered but others also available in different status then "in progress"
3. if nothing delivered but one submitted then "AWAITING PROGRESS"
4. otherwise print "AWAITING SUBMISSION"

--> If then else means --> CASE

The screenshot shows a SQL query result for 'select * from Customer_Order;'. The table has columns: CUSTOMER_NAME, ORDER_ID, and STATUS. The data is as follows:

CUSTOMER_NAME	ORDER_ID	STATUS
John	J1	DELIVERED
John	J2	DELIVERED
John	J3	DELIVERED
David	D1	SUBMITTED
David	D2	DELIVERED
David	D3	CREATED
Smith	S1	SUBMITTED
Smith	S2	CREATED
Smith	S3	SUBMITTED
KRISH	K1	CREATED
Avnee	A1	DELIVERED
Avnee	A2	DELIVERED
Avnee	A3	DELIVERED
Avnee	A4	SUBMITTED
Sruti	SR1	SUBMITTED
Sruti	SR2	CREATED
Sruti	SR3	SUBMITTED
Michael	M1	DELIVERED

Handwritten annotations on the screenshot:

- Red: "All delivered so, → completed" (pointing to John's orders)
- Orange: "→ 1 Created then in progress" (pointing to David's orders)
- Green: "→ 3 delivered but 1 submitted then awaiting submission" (pointing to Avnee's orders)

- There are many ways of writing this sql. try it and let me know the trick.. we discussed in class.. we know the tricks
- Just let me know what trick we will use. sql is all about trick. just read the problem is english and translate in sql.

Method1 :

Select Customer_Name, Listagg(distinct Status,',') within Group (Order by Status) Status from Customer_Order Group By Customer_Name

The screenshot shows a SQL query result for the query: 'Select Customer_Name, Listagg(distinct Status,',') within Group (Order by Status) Status from Customer_Order Group By Customer_Name'. The table has columns: CUSTOMER_NAME and STATUS. The data is as follows:

CUSTOMER_NAME	STATUS
Avnee	DELIVERED,SUBMITTED
David	CREATED,DELIVERED,SUBMITTED
Emmy	DELIVERED
Hanna	CREATED,SUBMITTED
Harris	CREATED,DELIVERED,SUBMITTED
John	DELIVERED
KRISH	CREATED
Michael	DELIVERED
Smith	CREATED,SUBMITTED
Sruti	CREATED,SUBMITTED
Zoe	CREATED

Handwritten annotation: "all data here" (pointing to the entire table)

Now, I have to go and check this sting. If that string has only DELIVERED and nothing else means : completed. if that string contains any things else then IN PROGRESS. we can use INSTR here as below:

With Report as (
Select Customer_Name, Listagg(distinct Status,',') within Group (Order by Status) Status from Customer_Order Group By Customer_Name

```

)
Select Customer_name, (Case When Instr(Status,'DELIVERED') > 0 And Instr(Status,',') = 0 Then 'COMPLETED'
When Instr(Status,'DELIVERED') > 0 And Instr(Status,',') > 0 Then 'IN PROGRESS'
When Instr(Status,'SUBMITTED') > 0 And Instr(Status,',') > 0 And Instr(Status,'DELIVERED') = 0 Then 'AWAITING PROGRESS'
Else 'AWAITING SUBMISSION'
End ) Final_Status
From Report;

```

With Report as (
Select Customer_Name, Listagg(distinct Status,',') within Group (Order by Status) Status **from** Customer_Order Group By Customer_Name
)
Select Customer_name, (Case When Instr(Status,'DELIVERED') > 0 And Instr(Status,',') = 0 Then 'COMPLETED'
When Instr(Status,'DELIVERED') > 0 And Instr(Status,',') > 0 Then 'IN PROGRESS'
When Instr(Status,'SUBMITTED') > 0 And Instr(Status,',') > 0 And Instr(Status,'DELIVERED') = 0 Then 'AWAITING PROGRESS'
Else 'AWAITING SUBMISSION'
End) Final_Status
From Report;

Query Result x Query Result 2 x Query Result 3 x

SQL | All Rows Fetched: 11 in 0.009 seconds

	CUSTOMER_NAME	FINAL_STATUS
1	Avnee	IN PROGRESS
2	David	IN PROGRESS
3	Emmy	COMPLETED
4	Hanna	AWAITING PROGRESS
5	Harris	IN PROGRESS
6	John	COMPLETED
7	KRISH	AWAITING SUBMISSION
8	Michael	COMPLETED
9	Smith	AWAITING PROGRESS
10	Sruti	AWAITING PROGRESS
11	Zoe	AWAITING SUBMISSION

✓ output

Method2: Use exists

```

select distinct C.Customer_name,
case when STATUS = 'DELIVERED' and Not Exists (select 1 from customer_order c1 where c1.customer_name = C.customer_name and status <>
'DELIVERED')
then 'completed'
when STATUS = 'DELIVERED' and Exists (select 1 from customer_order c1 where c1.customer_name = C.customer_name and status <> 'DELIVERED')
then 'in progress'
when STATUS <> 'DELIVERED' and Exists (select 1 from customer_order c1 where c1.customer_name = C.customer_name and status = 'SUBMITTED')
then 'AWAITING PROGRESS'
Else 'AWAITING SUBMISSION'
end as final_status
from customer_order c

```

Method3: Use count

```

select C.*,
case when STATUS = 'DELIVERED' and 0 = (select count(*) from customer_order c1 where c1.customer_name = C.customer_name and status <>
'DELIVERED')
then 'completed'
when STATUS = 'DELIVERED' and 0 < (select count(*) from customer_order c1 where c1.customer_name = C.customer_name and status <>
'DELIVERED')
then 'in progress'
end as final_status
from customer_order c

```

--> Try any method, any way, just make your translation.